The ultrasonic vibrations produced by generator spread the oil and transfer the vibration to the vessel with mercury in water.

(iv) Peptisation:

By addition of suitable electrolytes, precipitated particles can be brought into colloidal state. The process is termed as peptisation and the electrolyte added is called peptising or dispersing agent.

$$\underset{\text{Precipitate}}{AgCl} \xrightarrow{HCl} \underset{\text{Colloid}}{AgCl}$$

2) Condensation Methods:

When the substance for colloidal particle is present as small sized particle, molecule or ion, they are brought to the colloidal dimension by condensation methods. Here care should be taken to produce the particle with colloidal size otherwise precipitation will occur. Various chemical methods for the formation of colloidal particles.

(i) Oxidation:

Sols of some non metals are prepared by this method.

(a) When hydroiodic acid is treated with iodic acid, I_2 sol is obtained.

$$HIO_3 + 5HI \rightarrow 3H_2O + 3I_2(Sol)$$

(b) When O_2 is passed through H_2Se, a sol of selenium is obtained.

$$2H_2Se + O_2 \rightarrow 2H_2O + 2Se(sol)$$

(ii) Reduction:

Many organic reagents like phenyl hydrazine, formaldehyde, etc are used for the formation of sols. For example: Gold sol is prepared by reduction of auric chloride using formaldehyde.

$$2AuCl_3 + 3HCHO + 3H_2O \rightarrow 2Au(sol) + 6HCl + 3HCOOH$$

(iii) Hydrolysis

Sols of hydroxides of metals like chromium and aluminium can be produced by this method.

For Example,

$$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$$

(iv) Double decomposition

For the preparation of water insoluble sols this method can be used.

When hydrogen sulphide gas is passed through a solution of arsenic oxide, a yellow coloured arsenic sulphide is obtained as a colloidal solution.

$$As_2O_3 + 3H_2S \rightarrow As_2S_3 + 3H_2O$$

(v) Decomposition

When few drops of an acid is added to a dilute solution of sodium thiosulphate, the insoluble free sulphur produced by decomposition of sodium thiosulphate accumulates into small, clusters which impart various colours blue, yellow and even red to the system depending on their growth within the size of colloidal dimensions.

$$S_2O_3^{2-} + 2H^+ \rightarrow \underset{sol}{S} + H_2O + SO_2$$

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